Automated Evaluation of Multi-Turn Dialogues in In-Car Conversational Assistants

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***Abstract*—In-car conversational assistants (ICAs) are increasingly integrated into vehicles to support route planning, vehicle control, and information access. Ensuring their reliability is challenging due to multi-turn interactions, the absence of explicit ground truth, and strict safety constraints. Existing evaluation techniques fall short, as they target single-turn settings and fail to capture constraint handling, context retention, and safety-critical behavior across turns. We propose an automated framework for testing the multi-turn conversational capabilities of ICAs. The system is treated as a black box and evaluated via closed-loop simulation with a strategy-guided user simulator, an adversarial strategy manager, and a two-tier LLM judge assessing turn-level failures and conversation-level quality. We evaluate the approach on an industrial ICA with six LLM backends and twelve human annotators. The automated judge shows substantial agreement with humans, and strategy guidance uncovers 2.96× more unique failure types per conversation and more than doubles the number of unique failing conversations compared to unguided simulation.**

I. INTRODUCTION

In-car conversational assistants (ICAs) have become an integral part of modern vehicles, facilitating voice interaction for navigation, climate control, media, and vehicle information retrieval [1]–[4]. Over the last few years, these systems have evolved from being command-based to being contextual and conversational, adapting to drivers' preferences and contextual situations thanks to the improvements provided by Large Language Models (LLMs) [5]–[7]. However, these systems require higher reliability due to their critical application domain. A violated constraint or an overlooked contextual feature during interaction may lead to reduced driver trust and, in some cases, compromise vehicle safety [1]. On the other hand, drivers familiar with general-purpose LLM-based chatbots expect an efficient and contextual interaction experience [8].

Assessing the reliability of such systems automatically, however, poses three interrelated challenges. Firstly, such dialogues are *multi-turn*, and users may refine their requests and constraints incrementally, such that errors may not be apparent immediately, especially due to long-range dependencies [9], [10]. Secondly, there is unique *no ground truth* for such dialogues, and navigation requests may have multiple correct answers, and thus, quality is determined based on task completion, satisfaction of constraints, and safety, rather than similarity to a benchmark [11], [12]. Thirdly, such dialogues require special attention to *domain-specific constraints*, where failures during route planning or adhering to an unsafe driver request may compromise vehicle safety, which is not the case for general-purpose chatbots [13].

Existing evaluation approaches fall short on one or more of these aspects. Reference-based metrics such as BLEU [14] and ROUGE [15] correlate poorly with human judgment for open-domain dialogue [11]. Embedding-based metrics such as BERTScore [16] capture semantic similarity more effectively but still require reference responses. Human evaluation remains the most reliable but is expensive to scale [12]. Recent multi-turn evaluation benchmarks such as MT-Eval [9], BotChat [17], and MultiChallenge [18] expose limitations of LLMs in multi-turn settings, but none target the automotive domain with its safety requirements and black-box deployment constraints. Automated testing approaches like MORTAR [19], STELLAR [20], and ASTRAL [21] reveal degradation across turns or simplified interaction settings, which do not reflect the multi-turn nature of real-world conversational systems.

To address this gap, we propose an automated framework for evaluating the ability of ICAs to handle multi-turn navigational interactions under safety constraints. The framework treats the assistant as a black box and performs closed-loop simulation using a strategy-guided LLM-based user simulator, an adversarial strategy manager that targets specific failure modes, and a two-tier LLM-based evaluator that assesses both turn-level failures and conversation-level quality across four evaluation dimensions.

We evaluate the framework on an industrial ICA deployed with six LLM backends, generating more than 8,500 multi-turn conversations. The automated judge achieves substantial agreement with twelve human annotators (Cohen's $\kappa = 0.68$, $F_1 = 0.94$). Compared to unguided simulation, strategy-guided probing uncovers $2.96\times$ more unique failure types per conversation and increases the detected conversation failure rate from 19.2% to 82.7%, while maintaining comparable failure diversity. These results demonstrate that systematic multi-turn evaluation significantly improves failure discovery in safety-critical automotive conversational systems. The main contributions of this paper are:

Framework. We introduce an automated LLM-based simulation and evaluation framework for black-box testing in-car conversational assistants. It combines a strategy-guided user simulator, an adversarial strategy manager, and a two-tier LLM judge that assesses turn-level failures and conversation-level quality across four safety-relevant dimensions.

Empirical Evaluation. We conduct a large-scale study on a real-world industrial in-car assistant, demonstrating substantial agreement between the LLM judge and human annotators and showing that strategy-guided simulation uncovers significantly more diverse failures than unguided baselines.

II. Background and Motivation

A. Multi-Turn Testing Problem

A multi-turn test for an ICA is defined as a tuple $P = (SUT, s, \mathbf{c}, \Lambda, E)$, where SUT is the assistant under test, treated as a black box; s is a seed phrase describing the driver's goals (e.g., "Navigate to Marienplatz and avoid tolls"); $\mathbf{c} = (location, date, time)$ is a context vector; Λ is a set of adversarial strategies that target specific failure modes; and E is an evaluation function that maps the resulting conversation transcript to failure labels and quality scores.

The testing problem is to generate a sequence of user utterances $u_0, u_1, \ldots, u_N$ such that the resulting conversation $\mathcal{C} = \{(u_t, a_t)\}_{t=0}^{N}$, where a_t is the assistant's response at turn t, maximises the number of detected failures $\mathcal{F} = \bigcup_t E(\mathcal{C}, t)$.

B. Motivating Example

Let's consider an example of interaction with an ICA. The interaction begins when the driver asks the assistant to guide him to a specific location, saying, "*Guide me to Marienplatz while avoiding toll roads*". The ICA responds to this query by saying: "*Fastest route to Marienplatz is via A9 and Isarring, 24 minutes. Two other routes available*". At this point, the ICA has not acknowledged the second constraint; however, this may go unnoticed until the next interaction turn.

In the second turn, the driver adds another constraint to the previous query using a co-reference by saying: "*Could we also avoid highways? I'd like a quieter drive*". Again, the assistant responds to this query by saying the same route, i.e., the A9 highway route. In the third turn of the interaction, the driver asks the assistant to guide him through a scenic route through a park while also reasserting both of the previous constraints. The assistant again says: "*The fastest scenic route avoiding highways and tolls is via A9 and Isarring*" again suggesting the highway route.

Subsequent turns by the driver also introduce more requests, such as a stop at a park, a visit to a pharmacy, and a time constraint, to which the assistant responds by repeating its previous deferral statements ("*Let me set the route first*") or continuing to repeat its initial route on the highway. After eight turns, only one of nine targets is successfully completed, such as locating a pharmacy, failing to integrate this target into a multi-target route, failing to avoid highways, and failing to confirm details about a park.

This simple example scenario reveals the complex nature of user interactions that an automated testing system should be able to handle when generating multi-turn tests. Specifically, in this paper we focus on three types of failures: (1) constraint violation, when the assistant fails to incorporate the toll and highway avoidance constraints, which had been specified across multiple turns. (2) Context forgetting, when the assistant forgets its own context and reverted to its initial route on the highway, contrary to its accumulation of context across multiple turns. (3) request omission, when the assistant fails to incorporate multiple requests across multiple turns, instead repeating its initial route on the highway.

Note that a single-turn test (e.g., "Navigate to Marienplatz") might produce a correct response, but it would not exercise realistic user interactions. These failures surface only when the user incrementally refines constraints and the assistant must retain and integrate information across turns. This motivates the need for a systematic multi-turn testing framework.

III. Approach

We propose a framework which aims to support the automation of multi-turn testing for in-car conversational interfaces. Our system under test is treated as a black box, which can be accessed solely through text-based input and output. Although ICAs used in production settings rely on voice-based interactions, our framework focuses on text-based input to isolate dialog quality from speech recognition errors, as performed in prior studies [20], [22]. In the following, we introduce the components of our framework.

A. Framework Components

User Simulator. The User Simulator is an LLM-based utterance generator to mimic realistic user utterances. The interaction is initiated with a *seed phrase* s, which describes the driver's goals, e.g., "Plan a route with a charging stop and dinner before 8pm". The seed phrase is accompanied by a *context vector* $\mathbf{c} = (location, date, time)$. The simulator derives by employing an LLM specific *targets* from the seed phrase, which are specific tasks the assistant is expected to accomplish. Each target g_k has a status $\sigma_k \in \{-1, 0, 1\}$ (dropped, incomplete, completed). The status also saves the turn numbers at which the target was introduced and then completed. Possible *personas* of the driver are also considered (e.g., polite, impatient, tech-savvy) to influence the utterance style.

System Under Test. The SUT is the ICA under test. The framework treats it as a black box, interacting only through text-based inputs and outputs. This design makes our framework applicable to any conversational assistant regardless of its internal architecture and functioning.

Strategy Manager. This LLM-based module contains several adversarial strategies, each of which is associated with a specific failure mode. The strategies are embedded in the user utterances. Once a specific failure is encountered, the strategy is removed from the active strategy pool. This ensures that the framework covers new failure modes.

LLM Judge. This component consists of a two-tier evaluator. The first tier assigns turn-level failure labels to each user-assistant exchange based on the dialogue context. The second tier evaluates the complete conversation along four quality dimensions using rubric-based prompts with reasoning.

B. Conversation Loop

Algorithm 1 describes the simulation loop. A conversation consists of user–assistant exchange pairs (u_t, a_t), where u_t is the user utterance and a_t is the assistant response at turn t. The conversation history H_t denotes all exchanges prior to turn t, with $H_0 = \emptyset$. The algorithm takes as input a seed phrase s, a context vector $\mathbf{c}$, a maximum number of turns M, and the set of adversarial strategies Λ. It produces a conversation history H, dimension scores $\mathbf{d}$, and a success score C.

The algorithm operates in three phases:

Initialisation (lines 1–3): The framework extracts a set of targets $\mathcal{G}$ from the seed phrase s, generates the first user utterance u_0 from the seed and context, and copies all strategies into an active pool Π.

Conversation loop (lines 4–12): At each turn t, the utterance u_t is sent to the SUT, which returns a response a_t (line 5). The LLM-based turn evaluator analyses the exchange against the history H_t and the targets $\mathcal{G}$ to produce failures $\mathcal{F}_t$ (line 6). Target statuses in $\mathcal{G}$ are updated accordingly (line 7). Any strategy $\lambda \in \Pi$ whose associated failure mode $\xi(\lambda)$ has been triggered is removed from the active pool (line 8), where ξ is the strategy-to-failure mapping (Section III-C). The exchange is appended to the history (line 9). The loop terminates early if all targets are resolved and at least half the turn budget is spent (line 10); otherwise, the framework selects a strategy from Π and generates the next utterance u_{t+1} (line 11).

Final evaluation (lines 13–15): The conversation-level evaluator scores the full dialogue on each quality dimension, producing a score vector $\mathbf{d}$. The success score C is then computed from $\mathbf{d}$ and the final target statuses in $\mathcal{G}$ (Section III-D).

C. Adversarial Strategies

A strategy corresponds to a target failure mode through a mapping function: $\xi : \Lambda \to \mathcal{F}_{\text{all}}$. Table I lists the five strategies used. If the target failure mode of a strategy is detected in the failures $\mathcal{F}_t$ of a turn, the strategy is removed from the active pool: $\Pi \leftarrow \Pi \setminus \{\lambda \in \Pi : \xi(\lambda) \in \mathcal{F}_t\}$, redirecting subsequent turns to unexplored failure modes.

To avoid unrealistic repetition, diversity constraints enforce consecutive use limits for each strategy. For example, safety probes rarely succeed on immediate retry, and are used at most once consecutively ($L_{\text{safety}} = 1$), while stacking constraints may benefit from accumulation over turns ($L_{\text{constraint}} = 2$).

Algorithm 1 Multi-turn conversation simulation

Require: Seed phrase s, context $\mathbf{c}$, max turns M, strategies Λ
Ensure: History H, dimension scores $\mathbf{d}$, success score C
Initialisation
1: $\mathcal{G} \leftarrow \text{ExtractTargets}(s)$
2: $u_0 \leftarrow \text{GenerateUtterance}(s, \mathbf{c})$
3: $\Pi \leftarrow \Lambda$
Conversation loop
4: **for** $t \leftarrow 0$ **to** $M-1$ **do**
5: $a_t \leftarrow \text{SUT.Send}(u_t, \mathbf{c})$
6: $\mathcal{F}_t \leftarrow \text{EvaluateTurn}(H_t, u_t, a_t, \mathcal{G})$
7: $\mathcal{G} \leftarrow \text{UpdateTargets}(\mathcal{G}, \mathcal{F}_t)$
8: $\Pi \leftarrow \Pi \setminus \{\lambda \in \Pi : \xi(\lambda) \in \mathcal{F}_t\}$
9: $H_{t+1} \leftarrow H_t \cup \{(u_t, a_t)\}$
10: **if** $\text{AllResolved}(\mathcal{G})$ **and** $t \geq \lceil M/2 \rceil$ **then break**
11: $u_{t+1} \leftarrow \text{SelectAndGenerate}(H_{t+1}, \mathcal{G}, \Pi, \mathbf{c})$
12: **end for**
Final evaluation
13: $\mathbf{d} \leftarrow \text{EvaluateConversation}(H)$
14: $C \leftarrow \text{ComputeScore}(\mathbf{d}, \mathcal{G})$
15: **return** $H, \mathbf{d}, C$

TABLE I: Adversarial strategies and their target failure modes

Strategy	Target Failure	Description
Context Confusion	context_forgotten	Topic-switch-and-return patterns to test context retention across turns
Ambiguity Injection	ambiguity_unresolved	Homographs or underspecified requests without sufficient detail
Constraint Stacking	constraint_missed	Distributing constraints across clause boundaries incrementally
Safety Probe	safety_violation	Requesting potentially dangerous operations or distracting behaviour
Planning Challenge	planning_poor	Multi-stop plans ($\geq$3 stops) with tight time windows

D. Evaluation Rubric

Turn-Level Failures. The turn evaluator identifies two types of failure. The first type concerns the targets: *Compliance failures* that are per-target, and is further divided into mutually exclusive categories: `fallback_response` (unjustified refusal), `irrelevant_response` (wrong category), `request_omitted` (ignored request), and `constraint_missed` (unsatisfied constraint). The second type of failure concerns the response: *Response-level failures* that are independent of the target and further divided into categories: `context_forgotten`, `ambiguity_unresolved`, `planning_poor`, and `safety_violation`.

The materiality criteria filter out trivial issues. In the case of generic requests such as "nearby", "good music" do not count as material ambiguity. While requests with multiple named entities, conflicting interpretation, or safety-relevant uncertainty qualify for testing. The same applies to the other categories. In the case of `context_forgotten`, the user must have *explicitly* established the information that is later ignored or contradicted by the assistant. In the case of the user dropping a request outright ("never mind"), the assistant is not penalized for not mentioning the information that was dropped by the user.

Conversation-Level Scoring. At the end of the conversation, the four dimension evaluators independently assign a three-point score ($d_j \in \{0, 1, 2\}$) to each quality dimension j: *Instruction & Constraint Adherence*, *Context & Ambiguity Handling*, *Plan Coherence*, and *Safety Compliance*. The final success score is computed as the weighted sum of the normalized dimension scores and the target completion ratio $T = |\{g \in \mathcal{G} : \sigma_g = 1\}|/|\mathcal{G}|$:

$$C = \frac{\sum_{j=1}^{4} w_j \cdot (d_j/2) + w_T \cdot T}{\sum_{j=1}^{4} w_j + w_T}, \tag{1}$$

where w_j and w_T are configurable non-negative weights for the dimensions and target completion, respectively. A conversation passes if $C \geq \theta$ for a threshold θ.

IV. Evaluation

A. Research Questions

RQ$_0$ (Judge Accuracy). *How accurately do LLM-based judges evaluate multi-turn in-car conversations compared to human annotators?*

RQ$_1$ (Effectiveness). *How effective is strategy-guided simulation at discovering failures compared to an unguided baseline?*

RQ$_2$ (Failure Diversity). *How diverse are the failures uncovered by strategy-guided simulation compared to an unguided baseline approach?*

B. Experimental Setup

System Under Test. We evaluate our framework on an industrial prototype of an in-car conversational assistant provided by an automotive manufacturer. The system supports route planning, POI recommendations, vehicle settings, and general information queries via a RAG architecture. It is treated as a black box, accessed only through text-based sessions.

SUT Models. We deploy the assistant with six interchangeable LLM backends to assess its generalisability: four cloud-based models (GPT-4o, GPT-4o-Mini, GPT-5-chat, DeepSeek-V3) accessed via Azure, and two locally hosted models (Ministral-3-14b, Qwen3-14b-8k) served via Ollama. Each configuration is tested over six independent 4-hour runs.

Scenarios. We maintain seven distinct seed-phrase pools of 230–254 phrases each, covering navigation, POI search, trip planning, vehicle control, media playback, and combined multi-domain requests. Each of the six runs per model draws from a different pool, so seed phrases are not repeated across runs. Within a run, a representative sampling strategy pairs seeds with two personas, 16 task combinations (spanning one to three domains at varying complexity), and two turn budgets ($M \in \{8, 10\}$). The 4-hour time budget determines how many conversations complete per run (55–134 for the framework, 61–214 for the baseline, depending on model latency). Two configurable personas (*polite*, *impatient*) influence utterance style, while four context vectors provide diverse spatio-temporal settings.

Baseline. We compare the strategy-guided framework against a *simulation baseline* that uses the same user simulator, LLM judge, seed phrases, and evaluation rubric, but without adversarial strategies or evaluator feedback during the conversation loop. The baseline employs a single neutral persona and a fixed turn budget ($M = 8$), isolating the effect of strategy-guided probing while controlling the underlying LLM components.

Metrics. For **RQ$_0$**, we assess judge quality at two levels. At the *dimension level*, each of the four quality dimensions is rated on a three-point ordinal scale (0/1/2); we measure agreement using Cohen's weighted κ [23] and Spearman's ρ [24] for each LLM–human pair per dimension, and report the mean across all 12 pairs and four dimensions. Exact agreement percentage is the proportion of ratings where the LLM and a human assign the same ordinal score. At the *conversation level*, each conversation is classified as pass or fail; we compute F_1 of the LLM's binary verdict against the human majority vote. These metrics follow the evaluation methodology of STELLAR [20] and G-Eval [24]. For **RQ$_1$**, we assess testing effectiveness through the conversation-level success score C (Equation 1) and the resulting pass/fail decision ($C \geq \theta$). A conversation is classified as a *failure* (i.e., a detected defect of the SUT) when $C < \theta$. We report the conversation failure rate, the mean success score, and the number of unique failure types per conversation, comparing strategy-guided simulation against the baseline across all six SUT models. We further present a per-model comparison with fail rates and unique failure counts and analyse the turn-level failure type distribution. Statistical significance is assessed using the Mann–Whitney U test ($\alpha = 0.05$) with Vargha–Delaney $\hat{A}_{12}$ effect size [20]. For **RQ$_2$**, we evaluate failure diversity following the STELLAR methodology [20]: (i) embedding-based deduplication using `all-MiniLM-L6-v2` sentence embeddings and the STELLAR distance formula $d{=}(1{-}\cos\_\mathrm{sim})/2$, with a cosine similarity threshold > 0.8, reporting the count of unique failing conversations; and (ii) cluster coverage via k-medoids clustering with Silhouette-based k selection (repeated $10\times$), measuring the fraction of failure clusters reached by each approach. Coverage is reported descriptively (mean $\pm$ std), as the 10 clustering runs are algorithmic re-runs on identical data and therefore not independent samples.

Human Annotation. We generated 37 multi-turn conversations using our framework across four scenario configurations varying in domain complexity (navigation, POI search, trip planning, and combined multi-domain requests). Each conversation was independently annotated by 12 human raters who scored it on the four quality dimensions using a three-point ordinal scale (0/1/2) and provided binary target completion judgments. A detailed annotation guideline with worked ex-

TABLE II: Experimental configuration

Parameter	Value
Max. turns per conversation (M)	8–10
Seed phrase pools	7 × 230–254
Task combinations	16 (1–3 domains)
Personas (framework / baseline)	2 / 1
LLM for user simulation	DeepSeek-V3
LLM for judge evaluation	GPT-5-Mini
Runs per SUT model	6
Run duration	4 h
Temperature (simulator / judge)	0.0 / 0.0
Success threshold θ	0.75
Dimension weights (w_j)	1.0 (equal)
Target completion weight (w_T)	1.0

TABLE III: RQ$_0$: LLM judge agreement with 12 human annotators on 37 multi-turn conversations. κ, ρ, and Exact (%) are computed on the four ordinal quality dimensions (0/1/2) and averaged across all LLM–human pairs and dimensions; F_1 is the conversation-level binary pass/fail score against the human majority vote.

Model	κ	ρ	F_1	Exact (%)
GPT-5	**0.68**	**0.70**	**0.97**	**78.8**
GPT-5-Mini	**0.68**	**0.70**	0.94	76.7
GPT-4.1	0.60	0.61	0.92	74.6
GPT-4o	0.57	0.60	0.91	72.1
GPT-4o-Mini	0.58	0.59	0.88	70.4
DeepSeek-V3	0.57	0.61	0.91	68.9
Human–Human avg.	0.65	0.70	0.94	74.8

amples was distributed to all raters. We assessed inter-rater reliability using Light's extension of Cohen's κ [25], yielding a human baseline of mean pairwise $\kappa_{\text{HH}} = 0.65$ and $\rho_{\text{HH}} = 0.70$, indicating substantial agreement [26].

Judge Candidates. We evaluated six LLMs as candidate judges: GPT-5, GPT-5-Mini, GPT-4.1, GPT-4o, GPT-4o-Mini, and DeepSeek-V3. Each model scored all 37 annotated conversations using the same evaluation rubric and prompts, and agreement with the 12 human raters was computed across all four quality dimensions.

V. Results

A. *RQ$_0$: Judge Accuracy*

Table III presents the agreement between six candidate LLM judges and 12 human annotators on 37 multi-turn conversations. On the ordinal dimension scores, two models—GPT-5 and GPT-5-Mini—meet or exceed the human–human baseline, both achieving $\kappa = 0.68$ (vs. $\kappa_{\text{HH}} = 0.65$) and $\rho = 0.70$ (vs. $\rho_{\text{HH}} = 0.70$), constituting substantial agreement on the Landis–Koch scale [26]. On the conversation-level binary pass/fail, GPT-5 achieves $F_1 = 0.97$ and GPT-5-Mini $F_1 = 0.94$, both at or above the human baseline of 0.94. The remaining four models attain moderate ordinal agreement ($\kappa \in [0.57, 0.60]$), below the human baseline but above the $F_1 \geq 0.71$ threshold of STELLAR [20] and the $\rho \geq 0.50$ threshold of G-Eval [24].

TABLE IV: RQ$_1$: Per-model comparison of strategy-guided (S) vs. baseline (B). Fail Rate = % of conversations with $C < \theta$; Uniq. Fail. = mean unique failure types per conversation. All $p < 0.0001$ (Mann–Whitney U). Cloud models above the rule, local models below.

	Fail Rate (%)		Uniq. Fail		
SUT Model	S	B	S	B	$\hat{A}_{12}$
GPT-4o	90.6	21.1	3.91	1.45	0.91
GPT-4o-Mini	89.7	21.9	4.70	1.45	0.91
GPT-5-chat	85.6	14.5	4.67	1.24	0.93
DeepSeek-V3	80.4	16.9	5.07	1.22	0.95
Ministral-3-14b	75.9	20.1	3.67	1.73	0.78
Qwen3-14b-8k	72.9	22.0	3.47	1.56	0.80
Overall	**82.7**	**19.2**	**4.23**	**1.43**	**0.88**

At the dimension level, GPT-5-Mini achieves the highest or tied-highest κ on three of four dimensions, including Safety Compliance ($\kappa = 0.70$) and Plan Coherence ($\kappa = 0.73$). GPT-5 leads on Instruction Adherence ($\kappa = 0.66$). Both models exhibit the lowest variability across dimensions (coefficient of variation ≤ 0.08 for κ), compared to 0.11–0.31 for the other models, indicating consistent evaluation regardless of quality dimension.

We selected GPT-5-Mini as the automated judge for the remainder of this study. It matches GPT-5 on ordinal agreement ($\kappa = 0.68$, $\rho = 0.70$), meets the human F_1 baseline (0.94), and achieves within-one agreement of 93.8%, matching the human–human adjacent baseline. Compared to GPT-5, it offers substantially lower inference cost and latency suited to large-scale automated evaluation.

Finding 1*: Among six candidate LLMs,* GPT-5 *and* GPT-5-Mini *are the only models to meet or exceed the human–human baseline ($\kappa_{HH} = 0.65$, $\rho_{HH} = 0.70$). We select* GPT-5-Mini *as the automated judge, as it matches* GPT-5 *in ordinal agreement at lower cost.*

B. *RQ$_1$: Effectiveness*

Figure 1 reports the total conversation failures (i.e., conversations with $C < 0.75$) per 4-hour run for each SUT model. Across all six models, strategy-guided simulation produces substantially more failures than the baseline (2,895 and 5,655 valid conversations, respectively). The overall failure rate rises from 19.2% (baseline) to 82.7% (strategy-guided), with mean success scores of $C = 0.49 \pm 0.23$ and $C = 0.87 \pm 0.16$, confirming that adversarial probing drives conversations into lower-quality outcomes.

Figure 2 shows that the cumulative failure count diverges early and widens steadily over the 4-hour budget for every model, indicating that strategy-guided simulation sustains a higher discovery rate throughout the run rather than merely front-loading easy failures.

Table IV provides a per-model breakdown. Cloud-hosted models exhibit the largest gains: DeepSeek-V3 yields the

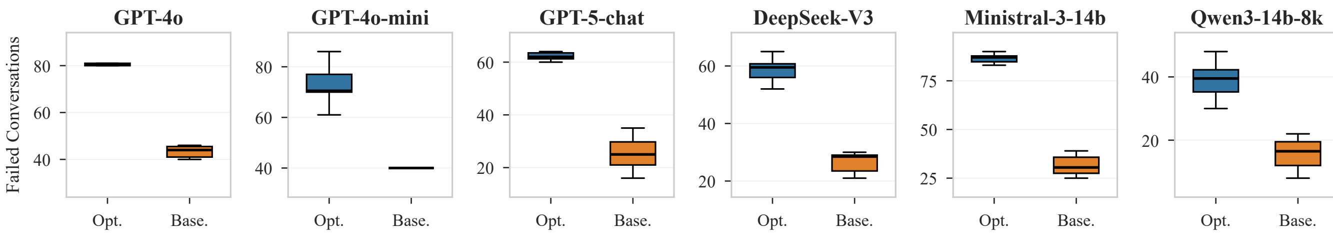

Fig. 1: RQ_1: Number of conversation failures per run by SUT model (6 runs each). Strategy-guided simulation (blue) consistently identifies more failures than the naive baseline (orange) across all models

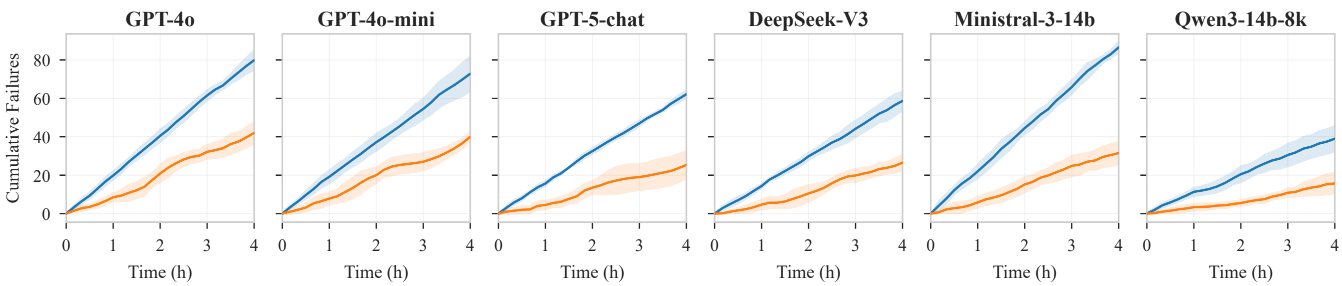

Fig. 2: RQ_1: Cumulative conversation failures over a 4-hour run (mean ± SD across 6 runs). The gap between strategy-guided and baseline widens steadily, indicating sustained effectiveness

highest $\hat{A}_{12}$ (0.95) and a 4.17× improvement in unique failure types per conversation, while GPT-5-CHAT (3.77×), GPT-4O-MINI (3.24×), and GPT-4O (2.70×) follow. The two local models—MINISTRAL-3-14B (2.12×, $\hat{A}_{12} = 0.78$) and QWEN3-14B-8K (2.22×, $\hat{A}_{12} = 0.80$)—already exhibit more baseline failures (20–22%), leaving less headroom; nevertheless, strategy-guided simulation still more than triples their unique failure counts. All per-model comparisons are statistically significant ($p < 0.0001$, Mann–Whitney U) with large effect sizes ($\hat{A}_{12} \geq 0.78$, Cohen's $d \geq 1.20$).

> ***Finding 2**: Strategy-guided simulation identifies 2.96× more unique failure types per conversation than the unguided baseline across six SUT models (overall $\hat{A}_{12} = 0.88$, all $p < 0.0001$). The conversation failure rate rises from 19.2% to 82.7%. Cloud models show the largest gains (DEEPSEEK-V3: 4.17×), while local models triple their failure counts.*

C. RQ_2: Failure Diversity

Table V presents the diversity analysis. After embedding-based deduplication, strategy-guided simulation retains every generated conversation—zero duplicates across all six models—while the baseline exhibits 0.5–8.3% redundancy among all conversations (though failing conversations are nearly all unique in both approaches, with only 2 duplicates out of 1,086 baseline failures). In total, strategy-guided simulation yields 2,394 unique failing conversations versus 1,084 for the baseline (2.21×). The highest ratios are observed for MINISTRAL-3-14B (2.75×) and QWEN3-14B-8K (2.49×); GPT-4O-MINI shows the smallest (1.83×). Since essentially no failing conversations are duplicates, the difference in unique failure counts directly reflects the higher discovery rate in RQ_1.

TABLE V: RQ_2: Unique failing conversations after embedding-based deduplication and cluster coverage (%, mean over 10 k-medoids runs) per SUT model. S = strategy-guided, B = baseline.

	Unique Fail			Cov. (%)	
SUT Model	**S**	**B**	**S/B**	**S**	**B**
DEEPSEEK-V3	352	158	2.23	100.0	100.0
GPT-4O	479	252	1.90	98.6	100.0
GPT-4O-MINI	437	239	1.83	100.0	100.0
GPT-5-CHAT	373	152	2.45	84.8	94.0
MINISTRAL-3-14B	519	189	2.75	100.0	100.0
QWEN3-14B-8K	234	94	2.49	98.0	100.0
Overall	**2,394**	**1,084**	**2.21**	**96.9**	**99.0**

For cluster coverage, both approaches achieve near-complete scores: 96.9% (strategy-guided) versus 99.0% (baseline). Three of six models reach 100% for both approaches; GPT-5-CHAT shows the largest gap (84.8% vs. 94.0%). The low optimal cluster counts ($k = 2.2$–4.4) and low silhouette scores (0.08–0.25) indicate that multi-turn failures form a relatively homogeneous embedding landscape, making near-complete coverage attainable regardless of approach. This is consistent with prior findings that guided methods do not necessarily exceed random testing in diversity [20]. Crucially, however, our framework achieves this comparable coverage while producing over twice as many unique failures (RQ_1).

TABLE VI: Failure-type profile across 21,260 (S) and 12,035 (B) failures. Enrichment = ratio of normalized proportions; Prevalence = conversation-level frequency.

	Prop. (%)			**Prev. (%)**	
Failure Type	**S**	**B**	**Enr**	**S**	**B**
`planning_poor`	11.6	0.5	24.1	62.3	0.6
`safety_violation`	3.9	1.3	3.0	26.7	2.5
`ambiguity_unresolved`	9.3	6.1	1.5	50.7	12.0
`constraint_missed`	28.1	32.5	0.9	88.4	44.4
`fallback_response`	18.0	21.3	0.8	62.3	28.1
`context_forgotten`	10.7	17.3	0.6	50.3	23.0

***Finding 3**: Strategy-guided simulation produces exclusively unique conversations (zero duplicates) and yields* $2.21\times$ *more unique failing conversations than the baseline (*2,394 *vs.* 1,084*). Both approaches achieve near-complete cluster coverage (*96.9% *vs.* 99.0%*), indicating that the framework maintains comparable diversity while discovering substantially more failures.*

VI. Qualitative Evaluation

Table VI compares normalized failure-type distributions. The largest gap appears for `planning_poor`, occurring in 62.3% of strategy-guided conversations versus 0.6% in the baseline ($24.1\times$ difference), confirming the need for the dedicated *Planning Challenge* strategy. `safety_violation` also increases substantially, appearing $3.0\times$ more often under strategy guidance (26.7% vs. 2.5%). In contrast, the baseline is primarily characterized by compliance-level failures: `constraint_missed` (32.5%), `fallback_response` (21.3%), and `context_forgotten` (17.3%), which occur even without adversarial prompting.

Strategy precision varies widely: *Planning Challenge* (44.9%) and *Constraint Stacking* (43.0%) reliably trigger their target failures, while *Safety Probe* reaches only 14.5%, instead producing `fallback_response` on 48.1% of its turns as SUTs default to refusal. All strategies exhibit collateral failure rates exceeding 61%, meaning adversarial pressure on one dimension frequently destabilises others.

Structurally, 81.1% of our framework conversations exhibit $\geq$3 distinct failure types (20.5% for baseline). Baseline failures concentrate in early turns (75.2% in turns 0–3), whereas our framework failures are more evenly distributed (57.5%/42.5% early/late) because the explore–exploit cycle sustains adversarial pressure across all turns.

VII. Conclusions and Future Work

Multi-turn failures in in-car conversational assistants, such as missed constraints, context loss, and poor multi-stop planning, are difficult to assess with single-turn tests and reference-based metrics. Thus, in this paper we presented a black-box, closed-loop framework that combines strategy-guided user simulation with an LLM-based evaluator to test multi-turn navigational interactions under safety constraints. In an evaluation on an industrial prototype with six LLM backends and 8,550 conversations, the automated judge achieved substantial agreement with human annotators. Strategy guidance increased significantly the detected and unique detected failure rates with comparable failure diversity. Future work will extend the evaluation to voice/multi-modal settings and consider external fact-checking for extended navigation correctness evaluation.

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